



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO NKK FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

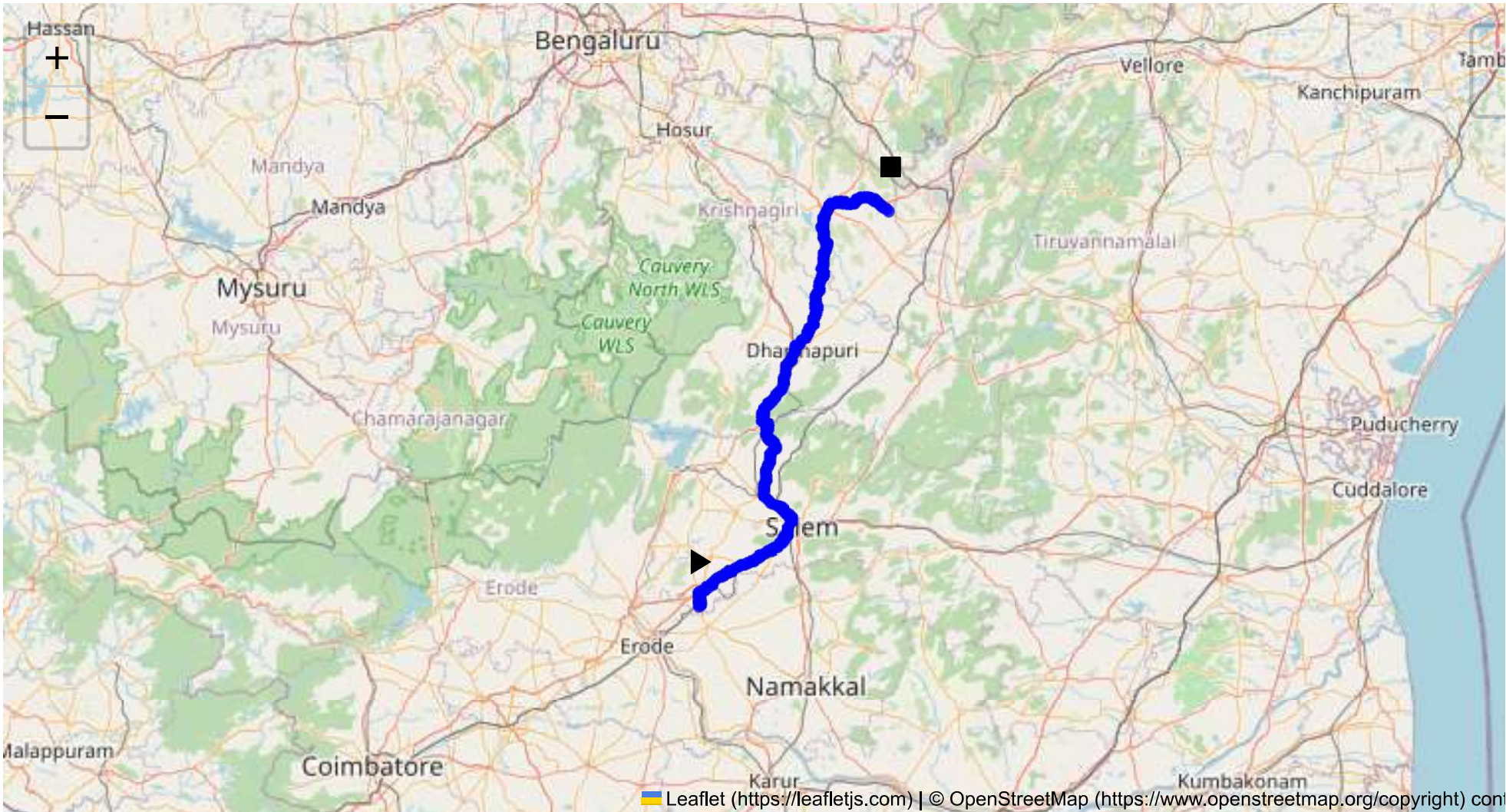
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 172.77 km
Estimated Duration: 3.1 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours
Start: (11.4381, 77.8734)
End: (12.512964, 78.396596)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Tamil Nadu 635104

1. Overview of the Route Map:

The route from CVQF+23W in Sangagiri, Tamil Nadu to G97W+C9C in Tamil Nadu spans approximately 172.77 kilometers, typically requiring around 3.13 hours for heavy vehicles. The route primarily traverses major highways and rural roads, facilitating connectivity between these regions.

2. Weather Conditions:

- The region experiences a tropical climate, with hot summers (March to May), a monsoon season (June to September), and mild winters (November to February).
- Monsoon season might lead to water-logged roads and reduced visibility. Drivers should prepare for slippery roads and potential flooding in low-lying areas.

3. Traffic Patterns:

- Traffic is generally dense during early morning (7:00 AM - 9:00 AM) and evening (5:00 PM - 7:00 PM) due to commuter flow.
- Congestion-prone areas include urban centers and intersections along the route.

4. Road Quality and Infrastructure:

- Generally, the major highways are well-maintained, but rural roads may suffer from potholes and uneven surfaces.
- Some sections could lack proper signage or lighting, increasing risk during night-time travel.

5. Alternative Routes for Emergencies:

- Alternative routes include using the NH-44 to divert around traffic or road closures. Routes via Salem can offer additional medical and mechanical facilities for unexpected contingencies.

6. Local Regulations on Hazardous Materials Transport:

- Transport of hazardous materials is regulated under the Motor Vehicles Act and enforces restrictions on times and designated routes.
- Ensure permits are obtained and adhere to weight restrictions and vehicle markings.

7. Historical Incidents:

- There have been incidents involving overturned trucks on rural roads due to sharp turns or inadequate road shoulders.
- Historically, monsoon-related disruptions have caused hazardous material spills.

8. Environmental Considerations:

- The route passes through agricultural areas; spills could have significant impacts on local ecosystems and communities.
- Drivers should be cautious near water bodies to prevent contamination in the event of an accident.

9. Communication Coverage:

- While major highways typically have good communication coverage, some rural stretches may experience dead zones.
- Utilization of satellite communication devices is advisable in such areas for maintaining contact.

10. Emergency Response Times:

- Emergency response teams may take between 30-60 minutes to reach remote areas.
- Proximity to urban centers like Salem reduces response times to 15-20 minutes in case of incident near these areas.

11. Overall Summary of Risk Assessment:

- The route is generally safe with proper adherence to road rules and in favorable weather conditions. However, risks increase significantly during monsoon months and peak travel times.
- The key hazards include weather-related disruptions and infrastructure limitations in rural areas.
- Preparation, including route familiarization, vehicle checks, and compliance with local transport regulations, is critical for safe transport of hazardous materials.

This assessment highlights the importance of planning and preparedness when transporting hazardous materials on this route, with a significant emphasis on weather and local road conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47986, 77.88397	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.62917, 78.11648	30 KM/Hr
14	Turn	High	11.64180, 78.11808	15 KM/Hr
15	Turn	High	11.64325, 78.11993	15 KM/Hr
16	Turn	Medium	11.67865, 78.12803	30 KM/Hr
17	Turn	Medium	11.68201, 78.12639	30 KM/Hr
18	Turn	Medium	11.69519, 78.10775	30 KM/Hr
19	Turn	Medium	11.73958, 78.05105	30 KM/Hr
20	Turn	Medium	11.75835, 78.05016	30 KM/Hr
21	Turn	Medium	11.86847, 78.08367	30 KM/Hr
22	Turn	Medium	11.87942, 78.08197	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
23	Turn	Medium	11.92585, 78.06507	30 KM/Hr
24	Turn	Medium	11.93319, 78.06419	30 KM/Hr
25	Turn	Medium	11.93557, 78.06103	30 KM/Hr
26	Turn	Medium	11.93704, 78.05402	30 KM/Hr
27	Turn	Medium	11.95664, 78.04781	30 KM/Hr
28	Turn	High	11.95818, 78.04978	15 KM/Hr
29	Turn	Medium	11.95988, 78.04948	30 KM/Hr
30	Turn	Medium	11.96304, 78.05136	30 KM/Hr
31	Turn	High	11.96399, 78.05135	15 KM/Hr
32	Turn	High	11.96580, 78.04923	15 KM/Hr
33	Turn	Medium	11.96675, 78.04939	30 KM/Hr
34	Turn	Medium	11.96791, 78.05070	30 KM/Hr
35	Turn	Medium	12.16983, 78.16769	30 KM/Hr
36	Turn	Medium	12.19881, 78.18157	30 KM/Hr
37	Turn	Medium	12.22459, 78.19136	30 KM/Hr
38	Turn	Medium	12.24731, 78.19601	30 KM/Hr
39	Turn	Medium	12.25060, 78.19977	30 KM/Hr
40	Turn	Medium	12.25329, 78.20039	30 KM/Hr
41	Turn	Medium	12.29110, 78.19909	30 KM/Hr
42	Turn	Medium	12.39904, 78.21764	30 KM/Hr
43	Turn	Medium	12.40156, 78.21845	30 KM/Hr
44	Turn	Medium	12.40589, 78.22275	30 KM/Hr
45	Turn	Medium	12.45154, 78.21792	30 KM/Hr
46	Turn	High	12.50709, 78.22230	15 KM/Hr
47	Turn	Medium	12.53142, 78.23923	30 KM/Hr
48	Turn	Medium	12.53353, 78.28848	30 KM/Hr
49	Turn	High	12.54834, 78.33419	15 KM/Hr
50	Turn	High	12.54807, 78.33430	15 KM/Hr
51	Turn	Medium	12.54349, 78.35367	30 KM/Hr
52	Turn	Medium	12.54371, 78.35566	30 KM/Hr
53	Turn	High	12.52973, 78.37500	15 KM/Hr
54	Turn	Medium	12.52553, 78.37498	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
10	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
11	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
13	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
18	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
19	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
20	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
23	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
24	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
26	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
27	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
30	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
31	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
32	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
36	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
41	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
48	hospital	Government Hospital, Bargur	12.5444, 78.357225	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
12	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
16	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
17	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
28	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
29	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
37	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
38	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
39	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
40	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
42	college	Government Polytechnic College	12.5352064, 78.2460463	30 km/h	Medium
43	college	Government Boys Arts and Science college	12.5323457, 78.2457274	30 km/h	Medium
44	college	IRT Polytechnique College	12.5354149, 78.2989061	30 km/h	Medium
45	college	P. S. V. College of Engineering and Technology	12.5350134, 78.3023583	30 km/h	Medium
46	school	Kingsley Garden Matriculation Higher Secondary School	12.5450463, 78.3151896	30 km/h	Medium
47	college	Government Engineering College	12.5442249, 78.331078	30 km/h	Medium
49	school	Goverment Girls Higher Sec School	12.5440902, 78.3581379	30 km/h	Medium
50	school	Concordia	12.537813, 78.3661043	30 km/h	Medium
51	school	IELC School For Blind	12.5391427, 78.3664826	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47986, 77.88397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62917, 78.11648



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64180, 78.11808



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67865, 78.12803



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68201, 78.12639



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93319, 78.06419



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93557, 78.06103



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95664, 78.04781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16983, 78.16769



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24731, 78.19601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40156, 78.21845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40589, 78.22275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.45154, 78.21792



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.50709, 78.22230



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.53142, 78.23923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.53353, 78.28848



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.54834, 78.33419



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.54807, 78.33430

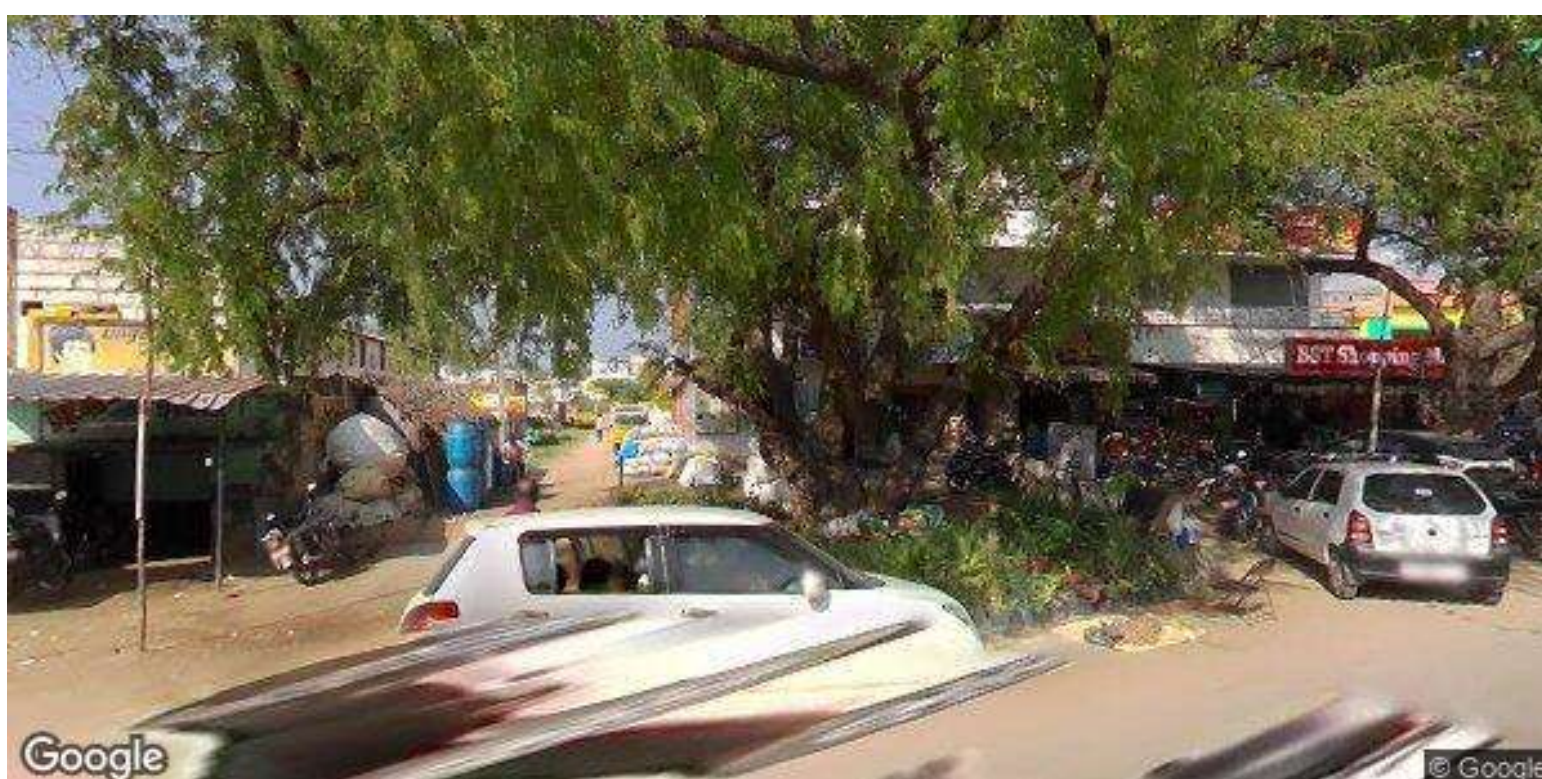


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.54349, 78.35367



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.54371, 78.35566



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.52973, 78.37500



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52553, 78.37498